

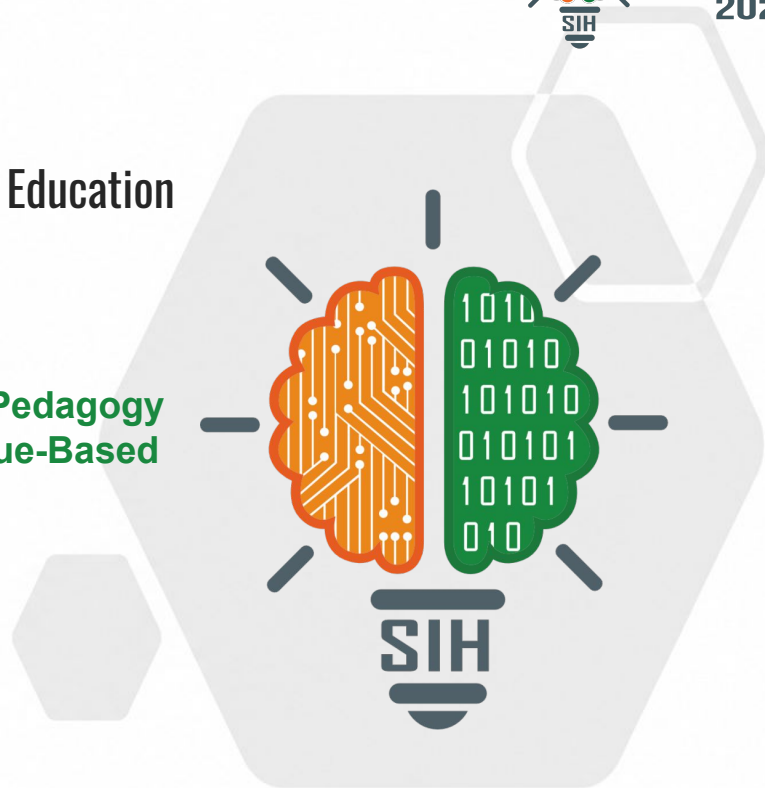
SMART INDIA HACKATHON 2026



SAMVAAD

Bridging the Language Gap in Tribal Primary Education

- Problem Statement ID: **SIH26042**
- Problem Statement Title: **AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education**
- Theme- **Smart Education**
- PS Category- **Software**
- Team ID- **N/A**
- Team Name – **Sankalp**



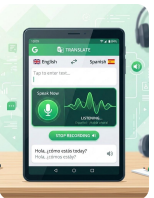
Bridging the Language Gap in Tribal Primary Education

PROBLEM EXISTING:



- Core Challenges in Scaling MTB-MLE**
How can Hindi-trained teachers effectively instruct tribal children when they lack the necessary linguistic tools?
- The Pedagogical Gap in Tribal Education**
Why do children in over 5,000 primary schools receive instruction in a language they cannot understand at home?
- Critical Barriers to Foundational Literacy**
What prevents scaling of Mother Tongue-Based Multilingual Education without a technology bridge?

SOLUTION PROPOSED:



- ✓ **AI-Powered Voice Translation:** Tailored specifically for low-cost Android devices to bridge Hindi and Santali in real-time.
- ✓ **Fine-Tuned IndicTrans2:** Customized and trained deeply on Hindi-Santali classroom and curriculum data.
- ✓ **Distilled & Quantized:** Employs INT8 quantization to compress models, enabling high-performance offline use on 2 GB RAM.
- ✓ **Low-Latency Pipeline:** Integrates offline ASR, Translation, and TTS targeting an end-to-end execution of < 3s.

(UVP) Focus Area	Existing Solutions	Our Unique Value Proposition
Target Use Case	Broad regional translation and general-purpose speech utility.	Purpose-built for interactive teacher-student classroom instruction.
Domain Accuracy	Struggles with educational pedagogy and specialized terminology.	Fine-tuned strictly on Hindi-Santali FLN curriculum and local terms.
Hardware Needs	Relies on persistent cloud access or heavy, uncompressed models.	Runs completely offline on low-cost ~2 GB RAM Android tablets.
Latency & Speed	Highly variable response times; unusable for rapid dialogue.	End-to-end voice loop benchmarked and guaranteed under 3s.
Language Strategy	Broad but extremely shallow support across multiple languages.	Deep specialization in Santali first; extensible to Ho and Mundari.

APP ARCHITECTURE

Offline Hindi → Santali Voice Translation for Classrooms

Bridging the Language Gap in FLN Classrooms



Hindi-Medium Teacher
(No Santali Training Required)

Provides Hindi Speech / FLN Content

Android Application (Completely Offline)

1 Input Processing

Teacher speaks in Hindi OR Type/Import FLN Text inputs

2 Hindi Speech Recognition (ASR) - Offline

Converts vocal inputs into normalized, structured Hindi text locally

3 Hindi → Santali Translation Engine

Fine-tuned IndicTrans2 model + context-aware translation pipeline

4 Santali Text-to-Speech (TTS)

Generates natural-sounding Santali speech outputs from translated text

5 Final Output Display & Playback

Displays Ol Chiki script while broadcasting audio translation to pupils

Performance

≤ 3 Seconds

End-to-End Latency

Device Target

~2 GB RAM

Low-Cost Androids

Content Focus

FLN Scope

Primary Curriculum

Connectivity

100% Offline

No Internet Needed

ON-DEVICE OPTIMIZATION



Model Compression

Knowledge Distillation



INT8 Inference

Quantized Engine Execution



ONNX Runtime

Accelerated Execution Layer



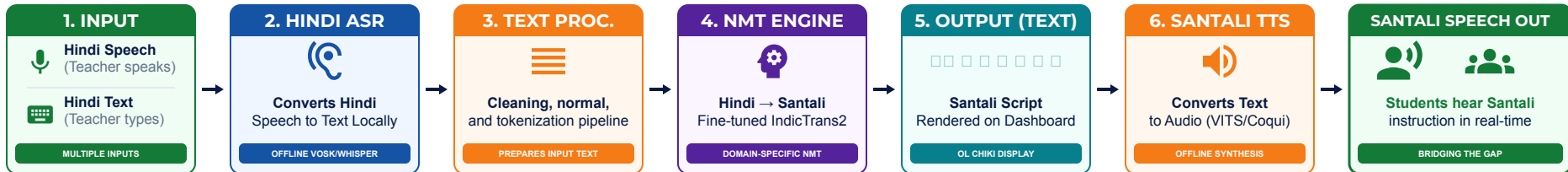
Model Caching

Minimized Cold Start Delay

Enabling Seamless Teacher-Student Communication

Accurate | Fast | Lightweight | Classroom-Ready

Runs Completely Offline on ~2 GB RAM Android Devices



DATASET & DATA PIPELINE



Our Dataset Will Include:

- ✓ Teacher Instructions (Classroom)
- ✓ Stories, Conversations, Phrases
- ✓ Subject Terms (Maths, EVS, etc.)
- ✓ FLN & MTB-MLE Vocabulary
- ✓ Questions & Answers
- ✓ Audio Data for ASR & TTS

Continuous expansion & improvement of the dataset.

MODEL DEVELOPMENT PIPELINE



Why this approach?

- ✓ **Fine-tuning:** Deeply adapts models for domain and classroom terminology.
- ✓ **Knowledge Distillation:** Reduces parameters and size while keeping performance.
- ✓ **INT8 Quantization:** Extremely low memory footprint and high CPU inference speed.

TECH STACK

AI/ML MODELS

IndicTrans2 (1B)
Base translation model
IndicTrans2 Distilled
Lightweight NMT
Vosk / Whisper Small
Offline Hindi ASR
Coqui TTS / VITS
Santali Speech Synth

LIBRARIES

PyTorch
Model training
Transformers
Hugging Face APIs
SentencePiece
Text tokenization
ONNX Runtime
Cross-platform exec

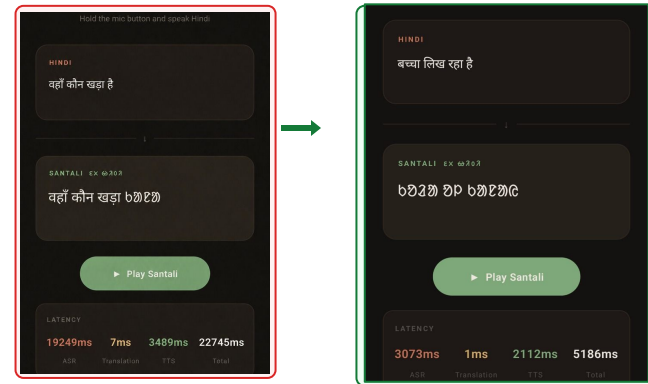
DEPLOYMENT

Android Studio
Kotlin base application
ONNX Mobile
On-device inference
Android NDK
C++ integration
SQLite / Prefs
Local DB and state

OPTIMIZATION & PERFORMANCE

- 🔧 **Stage-wise Latency Monitoring** (ASR → NMT → TTS)
- ⚡ **Model Quantization** (INT8 network execution)
- 🔧 **Memory & Inference Optimization** (local processing)
- 🔄 **Efficient Model Loading** and cold-start caching
- 🕒 **Target Pipeline Latency: ≤ 3 Seconds** end-to-end
- 📱 Optimized to run on ~2 GB RAM Android devices

PROTOTYPE DEMO — REAL PERFORMANCE



~22-Second End-to-End Latency

- ✗ Incorrect Santali output
- ✗ Excessively high pipeline delay
- ✗ No local compression used

~5.1-Second Measured Latency

- ✓ Highly accurate translation
- ✓ Optimized local INT8 execution
- ✓ Fully offline and secure

FROM ~22 SECONDS (INCORRECT) → ~5.1 SECONDS (ACCURATE)

A faster, lighter, and robust offline classroom prototype ready for field trial.

Measured result under real local trial workload.

FEASIBILITY

Technically Sound | Practically Implementable | Economically Viable

Technical

- IndicTrans2 + Offline Hindi ASR + Santali TTS
- Own Hindi-Santali classroom dataset
- Distillation + INT8 → ~2 GB RAM

~ 2 GB RAM



Innovation

- Built for MTB-MLE classrooms
- Hindi Speech → Santali Text → Santali Speech
- Target response ≤ 3 seconds



≤ 3 sec

Operational

- Fully offline on Android 9+
- Supports Hindi speech and text
- Models and resources stored locally



Economic

- Works on low-cost Android tablets
- Lightweight models reduce hardware



Low-cost
Android Tablets

VIABILITY

Social Impact | Policy Alignment | Long-term Potential

Market

- Hindi-medium teachers + Santali students
- Schools & educational programmes
- More accessible instruction



Policy & Revenue

- Supports PALASH MTB-MLE & FLN initiatives
- Suitable for schools and public education



PALASH
MTB-MLE

Investment & ROI

- Low infra dependency via on-device AI
- Open-source stack + low hardware needs



Lower Cost
Higher Reach

Scalability

- Santali as initial prototype language
- Same architecture supports Ho & Mundari
- Expanding dataset
- Modular ASR/TTS



TECHNICAL CHALLENGES

Key Risks and Mitigation Strategies

Risk

- Achieving ≤ 3-second response on ~2 GB RAM devices



Knowledge distillation + INT8 quantization

- Maintaining accuracy after model compression



Continuous benchmarking

- ASR and TTS memory / computation requirements



Optimize ASR → Translation → TTS individually

- Device constraints



Use lightweight Android-compatible runtimes



BUSINESS / USER CHALLENGES

Key Risks and Mitigation Strategies

Risk

- Teachers may not know Santali / Oi Chiki



Simple "Speak → Translate → Play" interface

- Classroom noise and different accents



Train and test using classroom-style data

- Possible translation errors in educational content



Show Santali text along with audio

- Need for accuracy in educational content



Validate translations with native Santali speakers

IMPACT AND BENEFITS

Transforming Tribal Classroom Communication Through Offline AI



Economic & Jobs

Affordable Deployment & Local Roles



Lower deployment cost

Designed for low-cost Android tablets with ~2 GB RAM.



Reduced infrastructure dependency

Entire classroom inference runs locally without cloud servers.



Local skill & job opportunities

Annotation, testing, and support create specialized roles.



Local AI ecosystem

Builds language tech capabilities and digital skills locally.

- Note:** Job creation is a potential long-term impact; the current roadmap focuses on technical enablement without specific headcount commitments.



Sustainability

Offline-First & Resource Efficiency



Offline operation

Classroom systems require zero persistent internet connection.



Low-resource models

Quantized INT8 & distilled configurations target low RAM.



Device life optimization

Extends usability and viability of older generation school hardware.



Digital learning content

Enables local rendering of OI Chiki script for lessons.

- Scope Note:** Designed specifically for hardware efficiency and access. Avoid drawing unsubstantiated direct carbon/e-waste reduction claims.



Social & Governance

Inclusive Mother-Tongue Learning



Language barrier resolution

Enables direct communication between teachers and Santali students.



Supports MTB-MLE policy

Directly drives mother tongue-based multilingual instruction.



FLN & Policy Alignment

Curriculum-aligned datasets focus on early literacy and numeracy.



Future Tribal Language Scaling

Core architecture designed for subsequent Ho and Mundari adaptation.

- Impact:** Validated alignment with foundational learning metrics and state-level tribal education programs.

Key Targets



Offline

No network req.



Low-Cost

~2 GB RAM target



Fast Response

Target ≤3s pipeline



Inclusive

Hindi → Santali



Privacy-First

100% Local voice



Scalable

Next: Ho & Mundari

1. AI & Translation Research

IndicTrans2 (AI4Bharat)

- Open-source multilingual NMT framework.
- Supports Hindi → Santali (sat_Olck) translation.
- Full baseline support for Santali's native OI Chiki script.
- Training & fine-tuning resources for low-resource Indic languages.

2. Speech Technology

Indic Parler-TTS (AI4Bharat)

- Multilingual Indian-language speech generation.
- Native, high-quality speech synthesis supporting Santali.
- Core foundation model for our offline voice stream synthesis.

3. Existing Government Initiatives

Adi Vaani (Ministry of Tribal Affairs)

- Standard government platform for tribal-language translation.
- Provides foundational speech-to-speech translation systems.
- Translates Hindi/English into Santali, Mundari, and Bhili.

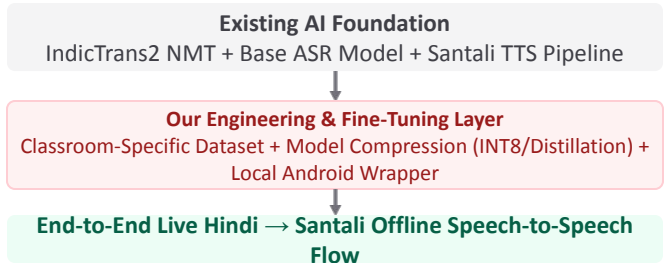
How Our Classroom Solution Stands Out

Feature / Domain	Standard Tech	Our Contribution
Translation Focus	General-purpose	Classroom Specialization
Interaction Type	Basic query-response	Teacher → Student flow
Dataset Base	Generic web corpora	Classroom FLN dataset
Deployment Size	Large server models	Distilled INT8 (~2 GB)
Architecture	Separate cloud API	End-to-End Local pipeline
Execution Mode	Cloud connectivity req.	100% Offline Android
Target Scale	Wide horizontal coverage	Santali → Ho & Mundari

Research Gap Addressed:

Bridging the leap from general server-side models to classroom-focused, highly resource-constrained, offline-first execution platforms optimized specifically for mother tongue-based multilingual education (MTB-MLE).

★ Research Gap / Contribution Flow



Reference Bibliography

[1] AI4Bharat — IndicTrans2

State-of-the-art open-source translation architecture supporting major low-resource Indian languages.

<https://github.com/AI4Bharat/IndicTrans2>

[2] AI4Bharat — Indic Parler-TTS

Hugging Face model repository for lightweight, native Indian accent text-to-speech synthesis pipelines.

<https://huggingface.co/ai4bharat/indic-parler-tts>

[3] Min. of Tribal Affairs — Adi Vaani

Government initiative details, speech translations, and programmatic frameworks for tribal linguistic development.

<https://adivaani.tribal.gov.in/>